

Gen-AI

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Unit1_Benchmark

A comparative analysis of transformer-based models was conducted to understand the impact of model architecture on task performance. Three models ,BERT, RoBERTa, and BART were evaluated across three NLP tasks: Text Generation, Fill-Mask, and Question Answering using Hugging Face pipelines.

The experiments highlighted that encoder-only models (BERT and RoBERTa) are unsuitable for free-form text generation, resulting in incoherent or limited outputs. In contrast, BART demonstrated relatively better performance due to its encoder–decoder architecture. Observations were documented and explained based on architectural differences, emphasizing task–model compatibility.

Unit1 Mini Project – AI Storyteller

An AI Storyteller application was implemented using the text-generation pipeline from Hugging Face. The system accepts a

starting sentence as input and generates a short creative story as output.

The project utilizes GPT-2 and DistilGPT-2, both decoder-only generative models, with non-deterministic sampling enabled to encourage creativity. Parameters such as `max_length`, `do_sample`, and `temperature` were used to control output length and randomness. This implementation demonstrates prompt-based text generation and compares the performance of a standard generative model with a lightweight alternative.